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Deleterious effects of roving on learned tasks

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ABSTRACT

In typical perceptual learning experiments, one stimulus type (e.g., a bisection stimulus offset either to the left or right) is presented per trial. In roving, two different stimulus types (e.g., a 30' and a 20' wide bisection stimulus) are randomly interleaved from trial to trial. Roving can impair both perceptual learning and task sensitivity. Here, we investigate the relationship between the two. Using a bisection task, we found no effect of roving before training. We next trained subjects and they improved. A roving condition applied after training impaired sensitivity.

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1. Introduction

In classical psychophysical experiments, one out of two stimulus alternatives is randomly presented per trial. For example, in a bisection task, two parallel lines are presented along with a central line that is offset either to the left or to the right (Fig. 1A). Subjects indicate the offset direction. In roving, one out of four stimulus alternatives (or even more) from two stimulus types is presented per trial, e.g., bisection stimuli separated by either 20' (arcminutes) or 30' with left and right offsets (Fig. 1A and B).

Roving hinders perceptual learning (Adini et al., 2004; Kuai et al., 2005; Otto et al., 2006; Yu, Klein, & Levi, 2004; Zhang et al., 2008), unless observers undergo abundant training, on the range of 18,000 trials (Parkosadze et al., 2008). This is roughly an order of magnitude greater than the 1500 trials that are sufficient for learning under non-roving conditions (Aberg & Herzog, 2009; Otto et al., 2006; Parkosadze et al., 2008). For sufficiently different stimuli, e.g., vertical versus horizontal bisection stimuli, roving does not hinder perceptual learning (Tartaglia, Aberg, & Herzog, 2009).

In a recent study, observers with and without experience in music-reading judged whether a dot was on or off a line on a musical staff (Wong et al., submitted for publication). The staff lines could be either horizontal or vertical. Music readers outperformed non-readers for the conventional horizontal staff lines

but not for vertical staff lines. Surprisingly, when vertical and horizontal staff lines were randomly interleaved from trial to trial (i.e., roving), experts were even worse than novices. It seems that roving affects perceptual learning *and*, in addition, sensitivity amongst experts, i.e., after a skill has been successfully learned.

Other studies, however, have found that expert sensitivity is unaffected by roving (Adini et al., 2004; Kuai et al., 2005; Nahum, Nelken, & Ahissar, 2012; Zhang et al., 2008). These studies used an assortment of tasks ranging from contrast increment detection (Kuai et al., 2005; Zhang et al., 2008) to auditory word discrimination (Nahum, Nelken, & Ahissar, 2012). Here, we investigated the effects of roving on pre- and post-training task sensitivity using bisection stimuli for which roving clearly affects learning.

2. General materials and methods

2.1. Observers

Observers included students, each of whom were from either the École Polytechnique Fédérale de Lausanne (EPFL) or from the University of Lausanne (UNIL), and who were naïve to the study's purpose. Ten observers participated in Experiment 1 and nine new observers in Experiment 2 (three females, mean age 22.81; and five females, mean age 23.6, respectively). Ten new observers participated in Experiment 3 (seven females, mean age 22.3) and another ten new observers participated in Experiment 4 (5 females, mean age 21.6). All observers had normal or corrected to normal acuity as assessed by the Freiburg visual acuity test (Bach, 1996). Observers were told that they could quit the

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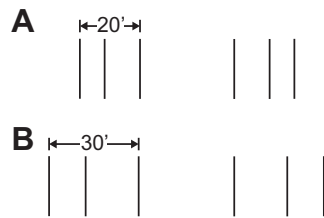


Fig. 1. (A) Per trial, a 20' bisection stimulus had its center line offset either to the left or to the right. The distance between the outer lines is 20'. (B) A 30' bisection stimulus. In roving, all four stimulus alternatives (A and B) were presented intermixed over trials.

experiment at any time they wanted and written informed consent was obtained. Observers were remunerated for participation (20 CHF per hour). All procedures conformed to the declaration of Helsinki.

2.2. Apparatus

Stimuli appeared on the center of either a Tektronix 608 display or a Hewlett Packard 1332A X-Y display, controlled by a PC via a 16-bit digital-to-analog converter (1 MHz pixel rate). Each observer was consistently tested with the same set-up at the EPFL. Line elements were composed of dots with a 200 μm pitch. The dot pitch was selected to make the dots slightly overlap, i.e. the dot size (or line width) was of the same magnitude as the dot pitch. Stimuli were refreshed at 200 Hz. Luminance was 80 cd/m^2 , as measured with a two-dimensional dot grid using the aforementioned dot pitch and refresh rate and a Minolta LS-100 luminance meter equipped with a close-up lens (Minolta No. 122). The room was dimly illuminated (0.5 lux) and background luminance on the screen was below 1 cd/m^2 . The viewing distance was 2 m.

2.3. Statistics

We measured sensitivity (d') as a function of training during the training sessions (Fig. 3A). To account for the different observers' improvement rates, we weighted our t -tests by learning strength, measured by the subjects' average improvement from the first four training blocks to the last four training blocks:

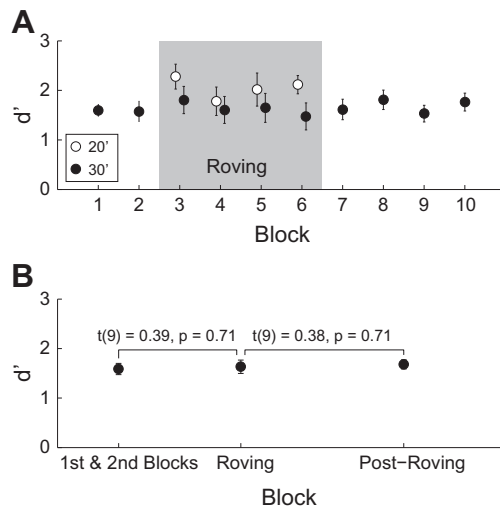


Fig. 2. Results for Experiment 1. (A) Black-filled symbols plot data for the 30' bisection stimulus and white-filled symbols plot data for the 20' bisection stimulus. The gray regions denote the roving blocks while the white regions are non-roving blocks. (B) Mean sensitivity averaged over the first two baseline blocks, the four roving blocks, and the four post-roving blocks for the 30' bisection stimulus. Symbols as in A. Error bars plot ± 1 SEM.

$$w_i = \frac{\overline{\text{last four}_i} - \overline{\text{first four}_i}}{\sum_{j=1}^n \overline{\text{last four}_j} - \overline{\text{first four}_j}} \quad (1)$$

$$D_i = w_i \cdot (\overline{\text{roving}_i} - \overline{\text{last four}_i}) \quad (2)$$

$$t = \frac{\bar{D}}{\sigma_D} \quad (3)$$

Here i and j index the observers, n is the total number of observers, $\bar{\cdot}$ denotes the mean and σ_D is the standard error on the difference scores. Under this formulation, in the case where subjects did not improve from their first four learning blocks to their last four, their weight w_i would be zero and their difference score would not count towards the resulting t -value. For the subject who improved the most from the first four training blocks to the last four training blocks, their weight would be the highest and their difference score would contribute the most to the resulting t -value. In this way, the t -statistic is un-biased by results from subjects who failed to learn the task, and for the remaining subjects, their contribution is weighted by how much they learned.

To investigate the influence of roving *after* training, in experiments 2, 3, and 4 we took the average sensitivity of all four roving blocks and subtracted the average sensitivity of the last four training blocks.

3. Experiment 1

In Parkosadze et al. (2008) it was shown that roving with bisection stimuli prior to learning does not affect bisection thresholds. Here, we replicated this effect with a slightly different procedure, showing that roving does not affect bisection *sensitivity* prior to learning.

3.1. Stimuli and task

Observers discriminated the offset of a central line (left or right) in a bisection stimulus. Bisection stimuli were 20' (arcminutes) tall (Fig. 1).

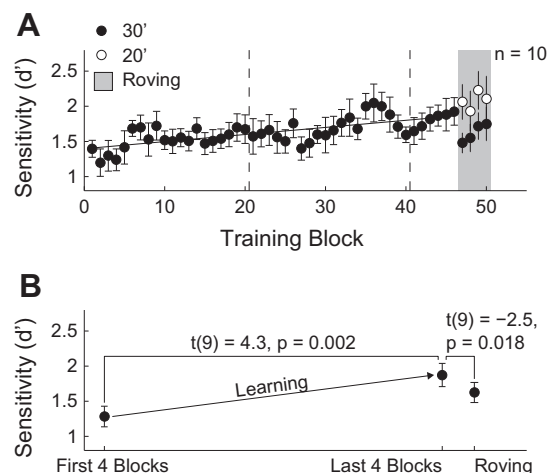


Fig. 3. Experiment 2. (A) Mean sensitivity (d') for eight observers. Black-filled symbols plot data for the 30' bisection stimulus and white-filled symbols plot data for the 20' bisection stimulus. The vertical dashed lines mark the different days. The gray shaded region marks the roving blocks. Performance improves from block 1 to block 46 with the 30' bisection stimulus. When in addition the 20' bisection stimuli are presented (roving) performance deteriorates for the 30' bisection stimulus. (B) Mean sensitivity averaged over the first four training blocks, the last four training blocks and the four roving blocks for the 30' bisection stimulus. Training led to a significant improvement. This improvement was diminished by post-training roving. Symbols as in A. Error bars plot ± 1 SEM.

First, subjects performed two blocks with a 30' wide bisection discrimination and then four blocks where the 30' wide bisection stimulus was roved with the 20' wide stimulus and an additional four blocks of just the 30' wide bisection stimulus. The experiment lasted for about 58 ± 2 min.

3.2. Results

Throughout all conditions, sensitivity remained unchanged (see Fig. 2; first two blocks versus roving: $t(9) = 0.39$, $p = 0.708$; roving versus post-roving: $t(9) = 0.38$, $p = 0.712$). Baseline thresholds measured prior to the learning experiment using the PEST procedure (Taylor & Creelman, 1967) were $59.07 \pm 8.12''$.

4. Experiment 2

In this experiment we test the hypothesis that roving affects performance on a learned task.

4.1. Methods

The stimuli and bisection task for Experiment 2 were the same as those for Experiment 1. On day one, observers first completed an initial threshold measurement using the PEST procedure (Taylor & Creelman, 1967) where we measured the offset threshold for the central line of the 30' bisection stimulus. In the subsequent training blocks, a fixed offset of $0.85 \times$ threshold was used for the 30' stimulus and this same value was also used for the 20' stimulus during roving. Each training block consisted of 80 trials. Observers completed 20 training blocks on day one (86 ± 3 min), another 20 training blocks on day 2 (82 ± 3 min) and six training blocks on day 3. Finally, on day 3, observers completed four blocks of 80 trials each where the 30' stimulus was randomly interleaved with the 20' stimulus (i.e., roving; threshold = $41 \pm 2'$).

4.2. Results

A one-way repeated measures ANOVA on the d' averages over the first four training blocks, the last four training blocks, and the roving blocks revealed a significant effect of block ($F(2,27) = 9.22$, $p = 0.001$, $\eta_p^2 = 0.51$). Post-hoc comparisons showed a significant difference between the last four training blocks and the first four training blocks ($t(9) = 4.35$, $p = 0.002$, Cohen's $d = 1.4$, large effect) and between the roving blocks and the last four training blocks ($t(9) = -2.53$, $p = 0.032$, Cohen's $d = 0.8$, large effect).

In accordance with previous studies, the sensitivity for the 20' bisection stimulus is higher than for the 30' bisection stimulus since the outer line distance is smaller and hence the task easier (Parkosadze et al., 2008). Performance in the 30' pre-training baseline was $78.51 \pm 4.69''$.

5. Experiment 3

The results of Experiment 2 leave open two important questions. First, does this effect hold up for more extensively trained tasks? It could be the case that the effect of roving decreases as the strength of learning increases. Second, does roving lead to unlearning or just to a deterioration during the roving conditions?

5.1. Stimuli and task

The stimuli and task for Experiment 3 were identical to those of Experiment 2 (Fig. 1), except that we added a third day of training in order to increase learning (20 blocks of 80 trials, 97 ± 2 min total), and we added four post-roving blocks (also 80 trials each).

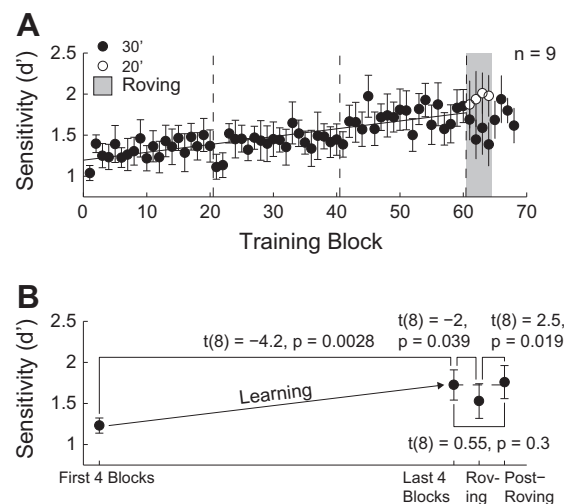


Fig. 4. Results for Experiment 3. (A) Mean sensitivity (d') for nine observers. (B) Mean sensitivity averaged over the first four training blocks, the last four training blocks, the four roving blocks and the four post-roving blocks for the 30' bisection stimulus. Training led to a significant improvement. This improvement was diminished by post-training roving, but recovered after roving. Symbols as in A. Error bars plot ± 1 SEM.

5.2. Results

We replicated the results of Experiment 2 (Fig. 4). Additionally, we found a significant difference between the roving blocks and the post-roving blocks ($t(8) = -2.47$, $p = 0.0192$, Cohen's $d = 0.825$, large effect), indicating that roving does not cause a prolonged deterioration of performance beyond the roving period.

Performance decrements in Experiment 3 following prolonged learning seem to be smaller than those observed in Experiment 2. To test this post-hoc hypothesis we conducted an independent measures t -test comparing the weighted differences between the last four training blocks and the roving blocks for Experiments 2 and 3. This test showed that the difference was not significant ($t(15) = 0.95$, $p = 0.1785$).

Baseline thresholds were comparable to those in Experiment 2 ($72.15 \pm 12.19''$).

6. Experiment 4

Post-training roving with the 20' bisection stimulus deteriorated sensitivity for the 30' bisection stimulus. Is this deterioration specific to stimuli that hinder perceptual learning? Here, we repeated Experiment 2, but mixed a horizontal bisection stimulus with a vertical bisection stimulus during roving, which does not hinder learning under roving conditions (Aberg & Herzog, 2009).

6.1. Results

Roving horizontal and vertical bisection stimuli has previously been shown not to hinder perceptual learning (Tartaglia, Aberg, & Herzog, 2009). Roving these stimuli after learning does not impede sensitivity for the vertical bisection stimulus (roving-last four: $t(9) = 1.41$, $p = 0.1$). Furthermore, after roving there is a significant post-roving depression (post-roving-roving: $t(9) = -3.41$, $p = 0.0039$, Cohen's $d = 1.079$, large effect) indicating that roving worsened post-roving sensitivity (Fig. 5). Post-roving sensitivity fell below the sensitivity for the last four training blocks (last 4-post-roving: $t(9) = 2.71$, $p = 0.0120$, Cohen's $d = 0.858$, large effect), indicating that the post-roving sensitivity decrement was not simply a return to baseline.

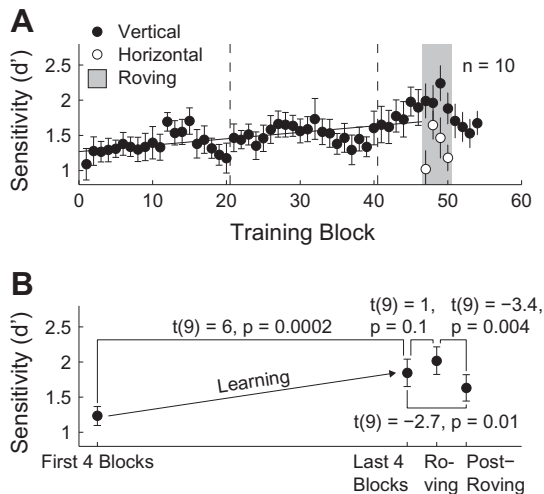


Fig. 5. Results for Experiment 4. (A) Black-filled symbols plot data for the vertical bisection stimulus and white-filled symbols plot data for the horizontal bisection stimulus. The vertical dashed lines mark the different days. The gray region marks the roving blocks. (B) Mean sensitivity averaged over the first four training blocks, the last four training blocks, the four roving blocks and the four post-roving blocks for the 30' bisection stimulus. Symbols as in A. Error bars plot ± 1 SEM.

Baseline thresholds in this experiment were comparable to those in Experiments 1, 2, and 3 ($77.63 \pm 16.98''$).

7. Discussion

First, we found that roving with 30' and 20' bisection stimuli did not affect novices' sensitivity. Second, training with 30' bisection stimuli improved performance as usual in perceptual learning. Third, post-training roving with the 30' and 20' stimuli deteriorated sensitivity for the 30' bisection task. Fourth, post-training roving of vertical with horizontal bisection stimuli (which do not impede perceptual learning) had no effect. Fifth, post-roving performance in this case was reduced.

We varied the amount of training between Experiments 2 and 3. In both experiments, we found that roving interfered with sensitivity, indicating that an increased number of trials does not protect against roving's deleterious effects. This is not surprising since similar perceptual learning studies with musical notation found that this long-lasting learning process is vulnerable to roving (Wong et al., submitted for publication).

Why did we find an effect of roving on post-learning sensitivity while others did not (Adini et al., 2004; Kuai et al., 2005; Nahum, Nelken, & Ahissar, 2012; Zhang et al., 2008)? All of these past studies adjusted task difficulty for each stimulus type individually, leading to similar performance levels for the roved tasks (Adini et al., 2004; Kuai et al., 2005; Nahum, Nelken, & Ahissar, 2012; Zhang et al., 2008). In the present study, the roved tasks differed in their difficulty levels (*thresholds* for the 20' stimuli are roughly 55% of the thresholds for the 30' stimuli; Parkosadze et al., 2008).

A recent mathematical analysis showed that roving hinders *learning* for tasks with differing difficulty levels because the critic of reinforcement learning models cannot assign "reward" individually to two similar stimulus types (because of the unsupervised bias; Frémaux, Sprekeler, & Gerstner, 2010; Herzog et al., 2012). This "unsupervised bias" argument is in line with the idea that roving occurs when stimulus types compete for resources in long-term potentiation (LTP; Aberg & Herzog, 2012) and also with the reverse hierarchy theory (RHT) positing that competition occurs between neurons in low-level (stimulus specific) areas that

are competing for wiring plasticity with higher-level areas (Ahissar & Hochstein, 2004; Ahissar et al., 2009). Regarding our present results, it might be that the unsupervised bias argument also applies to post-training sensitivity. Another possibility is that there was short-term learning in the post-learning blocks.

The unsupervised bias hypothesis, the RHT, and the LTP theories all predict that roving only hinders *learning* for tasks that recruit the same neural populations. For tasks that recruit differing neural populations there is no effect. Here, we found a similar result in a non-learning situation. Post-training interference by roving was restricted to "similar" horizontal bisection stimuli and did not occur for "dissimilar", orthogonal bisection stimuli. It seems that stimulus pairings that hinder perceptual learning also impair post-learning sensitivity. For pairings, where roving has no deleterious effect on learning, there is also no effect on post-learning performance (Fig. 5). It would be interesting to test whether perceptual learning of musical notes on a horizontal staff can be hindered by roving with musical notes on a vertical staff (Wong et al., submitted for publication).

Does roving cause long-lasting unlearning? The answer is that it depends on whether the roved stimuli cause interference. If we compare the results of Experiments 3 and 4 we observe a significant post-roving rebound effect. When sensitivity is impaired by roving (roving blocks in Fig. 4B), then post-roving sensitivity improves (relative to the roving blocks). Conversely, when sensitivity is not impaired by roving (roving blocks in Fig. 5B), then post-roving sensitivity deteriorates.

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